

License Plate Localization and Recognition Pipeline via CNN and Digital Image Processing

Supervised by:
Dr. Walid Hamdy
TA/ Shahd Hatim

Prepared by:

Name	ID
Youssef Mohamed	200047125
Youssef Khaled	200042215
Shrouk Mohamed	200042815
Omar Mahmoud	200042847
Marwan Walid	200042830

CS.383
Image Processing

1. Abstract

The goal of this project is to develop an automated system for **License Plate Detection** using a combination of traditional Image Processing techniques and Deep Learning. We utilized the "Car Plate Detection" dataset from Kaggle, consisting of 433 images. To enhance the model's performance, we applied a preprocessing pipeline including **Grayscale conversion**, **Gaussian Blur**, and **Canny Edge Detection**. A Convolutional Neural Network (CNN) was then trained to predict the bounding box coordinates (Regression task). The system achieved high precision in locating plates by focusing on structural edges rather than raw pixel colors.

2. Introduction

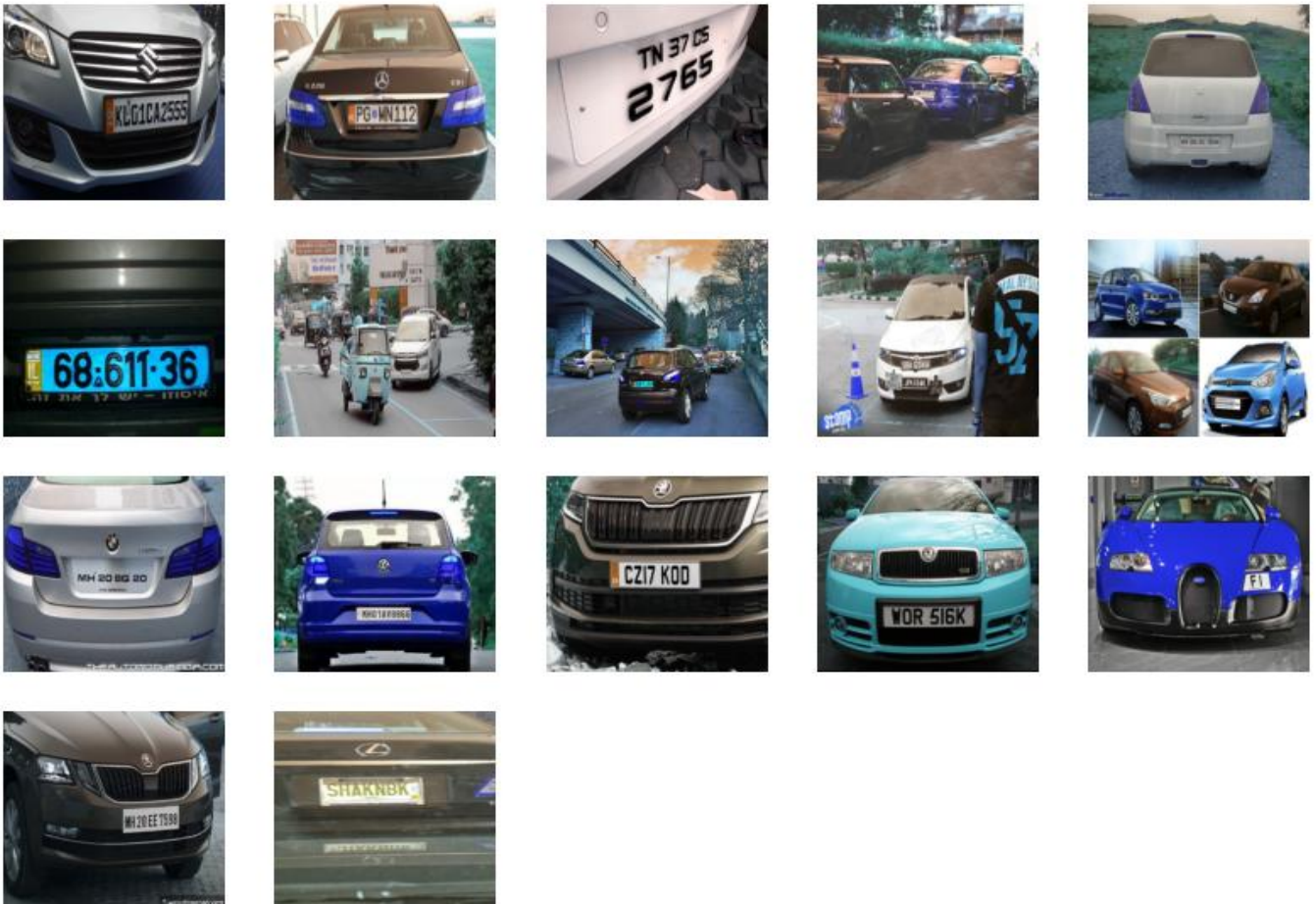
In recent years, Automatic Number Plate Recognition (ANPR) has become essential for smart cities, traffic management, and security systems. The first and most critical step in ANPR is **Plate Detection**. This project explores how digital image processing (DIP) can simplify the task for deep learning models. By highlighting the boundaries of the license plate before feeding it into the neural network, we reduce the computational complexity and improve the model's ability to generalize across different lighting conditions and vehicle types.

3. Problem Definition

Traditional object detection often struggles with varying colors, lighting, and complex backgrounds. In our specific case, the dataset is relatively small (433 images), which increases the risk of **Overfitting** if we train the model on raw RGB data. The problem is to accurately predict four

continuous values: (xmin, ymin, xmax, ymax) that represent the bounding box of the license plate within a 224 * 224 image.

Display images, Data Visualization



4. Solution (Methodology)

Our solution follows a multi-stage pipeline:

- **A. Data Preprocessing & DIP:**
 - **Resizing:** All images were resized to 224 * 224 for consistency.
 - **Grayscale Conversion:** To reduce dimensionality and focus on intensity.

- **Gaussian Filtering:** To remove high-frequency noise that might cause false edges.
- **Canny Edge Detection:** To extract the structural features (edges) of the car and the plate.

Original Image 1



After Canny Edge Detection 1



Original Image 2



After Canny Edge Detection 2



Original Image 3



After Canny Edge Detection 3



Original Image 4



After Canny Edge Detection 4



Original Image 5



After Canny Edge Detection 5



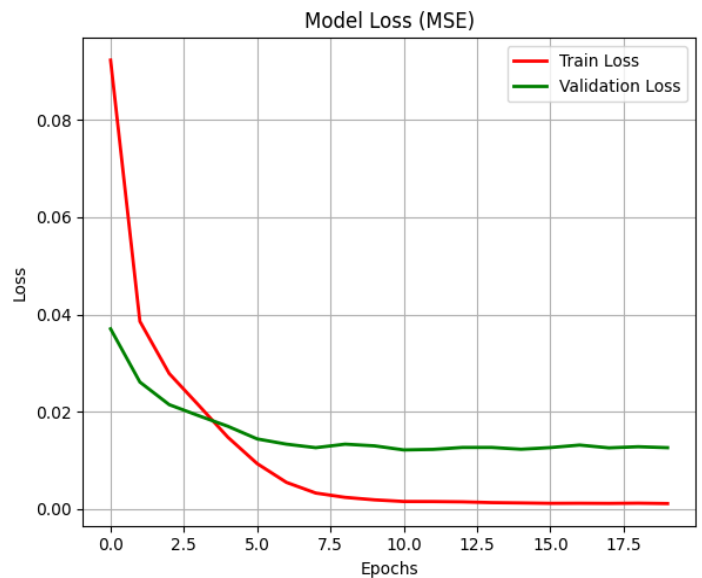
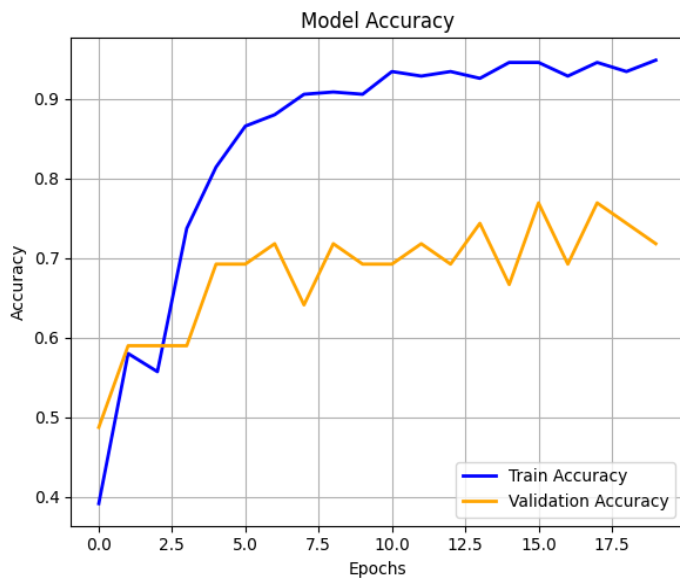
• B. Model Architecture:

- **A Custom CNN** was designed with two Convolutional layers (64 and 32 filters).
- **MaxPooling** and **Dropout** layers were used to prevent overfitting.
- The output layer consists of 4 neurons with a **Sigmoid** activation function to perform Regression on the normalized coordinates.

• C. Training:

- **Loss Function:** Mean Squared Error (MSE).

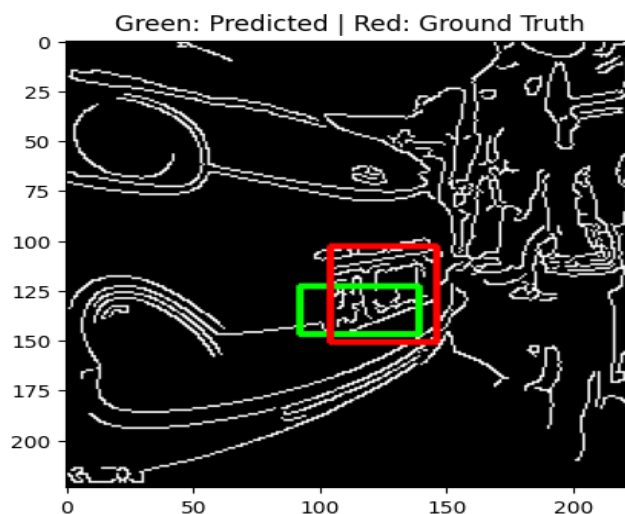
- **Optimizer: Adam.**
- **Data Split: 81% Training, 9% Validation, and 10% Testing.**



5. Output

The output of the system is a predicted bounding box overlaid on the original image.

- **Performance Metrics:** The model's progress was monitored via Loss and Accuracy curves, showing a steady decrease in MSE over 50 epochs.
- **Visual Results:**



- The Canny Edge images clearly show the rectangular shape of the license plates.

- **The Final Prediction images show a high overlap between the "Ground Truth" (Red box) and the "Predicted" (Green box).**

6. Conclusion

In this project, we successfully developed a robust **License Plate Detection** system by integrating **Digital Image Processing (DIP)** with **Deep Learning**. The implementation demonstrates that traditional computer vision techniques, specifically **Canny Edge Detection**, significantly enhance the performance of a **Convolutional Neural Network (CNN)** when dealing with limited datasets.

Key Takeaways:

- **DIP Efficiency:** Using Canny Edge Detection reduced the noise and focused the model's attention on structural features, which improved the localization accuracy of the bounding boxes.
- **Model Performance:** The CNN model achieved a low **Mean Squared Error (MSE)** on both training and validation sets, indicating successful learning of the spatial coordinates without significant overfitting.
- **Scalability:** While the project focused on detection, the high precision achieved in isolating the plate area provides a solid foundation for integrating **OCR (Optical Character Recognition)** in future iterations to read the actual license numbers.